Machined Part Write-Up

For the machined part, we focused on keeping the manufacturing process efficient and straightforward. The design allows it to be produced in three main orientations, ensuring minimal setup time and reducing the chances of error. By working on one primary surface per setup, we ensure that the holes and chamfers maintain their precision without needing excessive adjustments or tool changes.

1. **Setup and Drilling**:
   To simplify the drilling process, we used standard drill sizes of 6mm for the top holes and 9mm for the bottom holes. These sizes are common and ensure that the appropriate drill bits are easy to source. Using a 9mm drill bit for the inner fillet radius at 4.5mm was intentional—it allows for a single tool to handle the inner L-shape without requiring a separate step. This saves both time and effort. Additionally, the use of through holes (instead of blind holes) makes chip removal easier and improves accuracy.
2. **Chamfers and Edge Adjustments**:
   Chamfers replaced fillets to strengthen the part and simplify the machining process. The chamfers are positioned to keep a safe distance from the hole edges, reducing stress concentration and preventing cracks or deformation during use. The outer edges were kept at right angles to simplify the cutting path and minimize the complexity of the tool movements.
3. **Tooling and Tolerances**:
   We selected tools with appropriate depth-to-diameter ratios to ensure clean cuts without unnecessary wear on the machinery. Maintaining tight tolerances, especially for mating parts, was a priority. The design incorporates radii and chamfers that align with standard manufacturing practices to optimize the machining process while ensuring the final part's durability and functionality

Sheet Metal Part Write-Up

For the sheet metal part, our primary goal was to design a durable yet easy-to-produce part. We chose 3mm-thick steel for its strength and bendability. This thickness strikes the perfect balance between maintaining structural integrity and allowing smooth bends without cracking.

1. **Bend and Material Thickness**:
   The bends were designed around the 3mm thickness to ensure the material holds its shape well after forming. We accounted for the natural spring-back that occurs during bending by including a slight overbend in the design. The bend radii were carefully calculated to avoid fractures or warping while maintaining the material's structural properties. Generous bend radii and consistent bend dimensions throughout the part ensure uniformity and reduce strain on the tools.
2. **Cut and Form**:
   The flat pattern was designed to minimize material waste while maintaining precision. Laser cutting was selected as the primary method for shaping the sheet metal because it

allows for clean edges and tight tolerances. The pattern incorporates adjustments to account for stretching during bending, ensuring the final dimensions match the intended design. Proper nesting of the flat pattern reduces scrap and improves efficiency.

3. **Assembly and Finishing**:
   After forming, the edges are deburred to remove any sharp or uneven surfaces, ensuring safety during handling and use. The final part is checked for dimensional accuracy and surface finish to meet the design specifications. By following these steps, we ensure that the sheet metal part is both cost-effective and functional for its intended purpose.